

Unit 3: Data Collection, Sampling & Hypothesis Testing

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Data Collection: Primary vs. secondary sources, observation, questionnaires, and schedules Measurement & Scaling: Nominal, Ordinal, Interval, and Ratio scales, reliability, validity Sampling Fundamentals: Census vs. sample, sampling ...

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1. Methods of Data Collection

DEFINITION

Primary Data are gathered afresh and for the first time by the investigator, having an original character. **Secondary Data** are those already collected and compiled by an external agency, having passed through statistical processing.

1.1 Primary Data Collection Methods

- **Observation Method:** Data gathered through direct monitoring without asking the respondent.
 - *Structured vs. Unstructured:* Structured defines specific observational units in advance; Unstructured monitors naturally without predefined categories.
 - *Participant vs. Non-participant:* In participant observation, the researcher actively joins the group being observed.
- **Interview Method:** Oral verbal stimulus presented in face-to-face interaction or telephonically.
 - *Structured:* Uses predetermined, rigidly sequenced questions.
 - *Unstructured:* Flexible, conversational, exploring psychological nuances.
- **Questionnaires vs. Schedules:**
 - A *Questionnaire* is mailed or handed to respondents to be read and answered by themselves without assistance.
 - A *Schedule* contains identical questions but is read aloud and filled out by a trained enumerator/investigator who visits the respondent in person.

Criterion	Questionnaire Method	Schedule Method
Administration	Self-administered by the respondent.	Filled out by trained enumerator in person.
Cost & Coverage	Low cost, wide geographic reach.	High cost, requires hiring and training enumerators.
Non-Response Rate	High (many recipients discard forms).	Very low (enumerator ensures completion).
Literacy Dependency	Requires literate respondents.	Applicable even for non-literate respondents.

QUICK RECALL

Primary data is gathered afresh (observation, interview, questionnaires, schedules). Questionnaires are self-filled by respondents; schedules are administered by trained enumerators.

2. Measurement & Scaling Techniques

Measurement is the process of mapping numbers or symbols to properties of objects according to specific rules. S.S. Stevens established four levels of measurement scales:

Scale Level	Mathematical Properties	Permissible Statistics	Engineering Examples
1. Nominal	Identity / Classification only ($A = B$ or $A \neq B$). No order or distance.	Mode, Frequency, Percentage, Chi-square (χ^2).	Operating System type (Linux, Windows, macOS); Packet protocol (TCP, UDP, ICMP).
2. Ordinal	Classification + Rank Ordering ($A > B > C$). Intervals between ranks are unequal/unknown.	Median, Percentiles, Spearman's rank correlation.	Software defect severity (Low, Medium, High, Critical); Likert agreement scales (1 to 5).
3. Interval	Classification + Rank + Equal Intervals. Arbitrary (non-absolute) zero point. Addition & subtraction valid.	Mean, Standard Deviation, Pearson's r , t -test, F -test.	Temperature in Celsius/Fahrenheit; Calendar year (2025); Calendar time.
4. Ratio	Classification + Rank + Equal Intervals + Absolute True Zero . Ratios ($A = 2B$) are physically meaningful.	All statistics: Geometric mean, Harmonic mean, Coefficient of variation.	Execution latency (microseconds), Memory usage (megabytes), Clock frequency (GHz), Weight.

2.1 Tests of Sound Measurement

- **Reliability:** Consistency and repeatability of measurement. Assessed via Test-Retest reliability, Split-Half reliability, and Cronbach's Alpha ($\alpha \geq 0.70$).
- **Validity:** Whether the instrument truly measures what it purports to measure.
 - *Content Validity:* Adequate coverage of all facets of the construct.
 - *Criterion-Related Validity:* Correlation with an external benchmark (Predictive or Concurrent).
 - *Construct Validity:* Alignment with underlying theoretical concepts (Convergent & Discriminant).

QUICK RECALL

Stevens' 4 scales: Nominal (labels) → Ordinal (ranks) → Interval (equal units, arbitrary zero) → Ratio (equal units, absolute true zero). Ratio permits multiplication and division.

3. Sampling Fundamentals & Types of Sample Design

DEFINITION

A **Sample** is a finite subset of a population selected to estimate the parameters of that population. **Sampling Design** is the predetermined plan for obtaining a sample from a given population before any data are gathered.

3.1 Probability Sampling vs. Non-Probability Sampling

Category	Technique	Mechanism & Defining Feature
Probability Sampling (Every element has a known, non-zero inclusion probability)	Simple Random Sampling (SRS)	Every item has an equal probability ($\frac{1}{N}$) of selection via lottery or random number generators.
	Systematic Sampling	Select every k^{th} element from an ordered list after picking the first element randomly ($k = N/n$).
	Stratified Random Sampling	Population is partitioned into mutually exclusive, homogeneous strata (e.g. Freshmen, Sophomores, Seniors), and SRS is drawn from each stratum (Proportional or Disproportional).
	Cluster Sampling	Population is divided into heterogeneous naturally occurring clusters (e.g. server racks); a random sample of clusters is chosen, and all items in selected clusters are inspected.
Non-Probability Sampling (Selection based on subjective human discretion)	Convenience Sampling	Units chosen based on immediate geographic proximity or ease of access.
	Purposive / Judgment Sampling	Researcher hand-picks units deemed most representative based on professional expertise.
	Quota Sampling	Interviewer fills specific demographic quotas (e.g. 50 female developers, 50 male developers) non-randomly.
	Snowball Sampling	Existing study subjects recruit future subjects from among their acquaintances (used for rare/hidden populations).

QUICK RECALL

Probability sampling (SRS, Systematic, Stratified, Cluster) eliminates selection bias and supports statistical inference. Non-probability (Convenience, Judgment, Quota, Snowball) relies on subjective judgment.

4. Foundations of Hypothesis Testing

DEFINITION

A **Statistical Hypothesis** is an assertive, testable proposition concerning one or more unknown parameters of a population probability distribution.

4.1 Null Hypothesis (H_0) and Alternative Hypothesis (H_1)

- **Null Hypothesis (H_0):** The default hypothesis of "no difference", "no effect", or "no relationship" (e.g., $H_0 : \mu = 320$, or $H_0 : p = 0.20$). It is presumed innocent until proven guilty by compelling empirical evidence.
- **Alternative Hypothesis (H_1 or H_a):** The assertion that a real effect, difference, or relationship exists (e.g., $H_1 : \mu \neq 320$, $H_1 : \mu > 320$).

4.2 Decision Outcomes: Type I & Type II Errors

In evaluating a null hypothesis based on sample data, an investigator faces four possible outcomes:

True State of Nature	Decision: Accept (Fail to Reject) H_0	Decision: Reject H_0 (Accept H_1)
H_0 is Actually True	Correct Decision Probability = $1 - \alpha$ (Confidence Level)	Type I Error (α) "False Alarm" (Rejecting a true null hypothesis)
H_0 is Actually False	Type II Error (β) "Missed Detection" (Failing to reject a false null hypothesis)	Correct Decision Probability = $1 - \beta$ (Statistical Power of the Test)

Figure 3.1: Two-Tailed Hypothesis Testing Decision Regions ($\alpha = 0.05$)

Standardized Test Statistic (

Z) -1.96 Reject H_0 $\alpha/2 = 0.025$ Non-Rejection Region (Accept H_0) Confidence Level: $1 - \alpha = 0.95$ (95%) $+1.96$ Reject H_0 $\alpha/2 = 0.025$ 0

In a two-tailed test at $\alpha = 0.05$, if $|Z_{\text{cal}}| \geq 1.96$, the sample lies in the critical red region and H_0 is rejected.

QUICK RECALL

Type I Error (α) = false alarm (rejecting true H_0). Type II Error (β) = missed detection (failing to reject false H_0). Statistical Power = $1 - \beta$.

5. The Universal 5-Step Hypothesis Testing Protocol

Every hypothesis testing exam question must strictly follow these five structured steps:

1. **Step 1: State the Hypotheses:** Formulate H_0 (null condition of equality) and H_1 (alternative condition, noting whether one-tailed or two-tailed).
2. **Step 2: Specify the Significance Level (α):** Identify α (typically 0.05 or 0.01) and note the degree of confidence $1 - \alpha$.
3. **Step 3: Select and Calculate the Test Statistic:** Compute Z_{cal} , t_{cal} , χ_{cal}^2 , or F_{cal} using sample data and appropriate standard error formulas.
4. **Step 4: Determine the Critical Region (Rejection Threshold):** Read critical value (Z_{crit} or t_{crit}) from statistical tables for given α and degrees of freedom (df).
5. **Step 5: Formulate the Decision & Managerial Interpretation:**
 - If the test statistic falls in the critical region ($|Z_{\text{cal}}| \geq Z_{\text{crit}}$), **Reject H_0** .
 - Otherwise, **Fail to reject H_0** (Accept H_0). State the conclusion in plain language relative to the problem context.

QUICK RECALL

The 5 Steps: 1. State H_0, H_1 ; 2. Set α ; 3. Compute test statistic; 4. Find critical threshold; 5. Conclude (Reject or Fail to Reject H_0).

6. Parametric Tests (ZZZ-Test & Student's ttt-Test)

6.1 Z-Test for Population Mean (Large Sample $n \geq 30$ or Known σ)

FORMULA 1: STANDARD ZZZ-TEST FOR MEAN

$$Z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

where $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ for an infinite (or very large) population.

FORMULA 2: FINITE POPULATION CORRECTION FACTOR (FPC)

When sampling without replacement from a finite population of size N such that the sampling fraction exceeds 5% ($\frac{n}{N} > 0.05$), the standard error **must** be multiplied by the Finite Population Correction (FPC):

$$\sigma_{\bar{X}} = \left(\frac{\sigma}{\sqrt{n}} \right) \cdot \sqrt{\frac{N-n}{N-1}}$$

6.2 Z-Test for Population Proportion

FORMULA 3: ZZZ-TEST FOR SINGLE PROPORTION

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 \cdot q_0}{n}}}$$

where p_0 is the hypothesized proportion, $q_0 = 1 - p_0$, and $\hat{p} = \frac{x}{n}$ is the sample proportion of successes.

6.3 Student's t -Test for Small Samples ($n < 30$, Unknown σ)

FORMULA 4: ONE-SAMPLE STUDENT'S TTT-TEST

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}, \quad df = n - 1$$

where s is the sample standard deviation calculated with Bessel's correction ($n - 1$):

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

QUICK RECALL

Use Z -test for large samples ($n \geq 30$) or known σ . If finite population fraction $n/N > 0.05$, apply FPC $\sqrt{\frac{N-n}{N-1}}$. Use Student's t -test for small samples ($n < 30$) with unknown σ ($df = n - 1$).

7. Chi-Square (χ^2) Test of Independence

DEFINITION

The **Chi-Square (χ^2) Test of Independence** is a non-parametric statistical test used to determine whether a significant association exists between two categorical variables arranged in an $r \times c$ contingency table.

FORMULA: CHI-SQUARE TEST STATISTIC

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad df = (r - 1)(c - 1)$$

where O_{ij} is the observed frequency in cell (i, j) , and E_{ij} is the theoretical expected frequency under the null hypothesis of independence:

$$E_{ij} = \frac{(\text{Row } i \text{ Total}) \times (\text{Column } j \text{ Total})}{\text{Grand Total } N}$$

VALIDITY CONDITIONS FOR CHI-SQUARE

- Observations must be independent and collected randomly.
- Total sample size N should be reasonably large ($N \geq 50$).
- No expected cell frequency E_{ij} should be less than 5. If $E_{ij} < 5$, adjacent categories must be pooled or Yates' correction for continuity applied for 2×2 tables.

QUICK RECALL

$\chi^2 = \sum \frac{(O-E)^2}{E}$ with $df = (r - 1)(c - 1)$. Expected frequency $E = \frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}}$. Valid only when all $E \geq 5$.

8. Analysis of Variance (ANOVA) & Covariance (ANOCOVA)

8.1 One-Way Analysis of Variance (ANOVA)

While the t -test compares two means, **ANOVA** tests the equality of $k \geq 3$ population means simultaneously without inflating the Type I error rate:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k \quad \text{vs.} \quad H_1 : \text{At least one mean differs}$$

ANOVA partitions total variance into two orthogonal components:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{SS_{\text{Between}}/(k - 1)}{SS_{\text{Within}}/(N - k)}$$

8.2 Analysis of Covariance (ANOCOVA)

ANOCOVA combines ANOVA with linear regression. It removes the confounding effect of one or more uncontrolled continuous concomitant variables (covariates, X) that correlate with the dependent variable (Y), thereby reducing residual experimental error and sharpening treatment comparisons.

QUICK RECALL

ANOVA compares $k \geq 3$ means using the F -ratio ($MS_{\text{Between}}/MS_{\text{Within}}$).
ANOCOVA integrates regression to statistically adjust for confounding continuous covariates.

9. Solved University Exam Numericals

SOLVED EXAM PROBLEM 3.1: FINITE POPULATION MEAN TEST (SEE FEB/MAR 2025, Q5B, 10 MARKS)

Problem Statement: Suppose we are interested in a population of 20 industrial units of the same size, all of which are experiencing excessive labour turnover problems. The past records show that the mean of the distribution of annual turnover is 320 employees, with a standard deviation of 75 employees. A sample of 5 of these industrial units is taken at random which gives a mean of annual turnover as 300 employees. Is the sample mean consistent with the population mean? Test at 5% level of significance.

Step 1: Formulate Hypotheses:

- Null Hypothesis $H_0 : \mu = 320$ (Sample mean is consistent with population mean turnover).
- Alternative Hypothesis $H_1 : \mu \neq 320$ (Two-tailed test).

Step 2: Identify Parameters & Check Finite Population Condition:

- Population size $N = 20$
- Sample size $n = 5$
- Hypothesized mean $\mu_0 = 320$, Population standard deviation $\sigma = 75$
- Sample mean $\bar{X} = 300$, Significance level $\alpha = 0.05$

Check sampling fraction: $\frac{n}{N} = \frac{5}{20} = 0.25 = 25\% > 5\%$. Therefore, the **Finite Population Correction (FPC)** factor is strictly required!

Step 3: Calculate Standard Error with FPC:

$$\text{FPC} = \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{20-5}{20-1}} = \sqrt{\frac{15}{19}} \approx 0.8885$$

$$\sigma_{\bar{X}} = \left(\frac{\sigma}{\sqrt{n}} \right) \times \text{FPC} = \left(\frac{75}{\sqrt{5}} \right) \times 0.8885 = 33.541 \times 0.8885 \approx 29.8020$$

Step 4: Compute Test Statistic:

$$Z_{\text{cal}} = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}} = \frac{300 - 320}{29.8020} = \frac{-20}{29.8020} \approx -0.6711$$

Step 5: Decision & Conclusion:

For a two-tailed test at $\alpha = 0.05$, the critical value is $Z_{0.025} = \pm 1.96$.

Since $|Z_{\text{cal}}| = 0.6711 < 1.96$, the test statistic falls comfortably in the non-rejection region.

SOLVED EXAM PROBLEM 3.2: ZZZ-TEST FOR POPULATION PROPORTION (SEE FEB/MAR 2025, Q6B, 10 MARKS)

Problem Statement: The null hypothesis is that 20 per cent of the passengers go in first class, but management recognizes the possibility that this percentage could be more or less. A random sample of 400 passengers includes 70 passengers holding first class tickets. Can we reject the null hypothesis at the 5% level of significance?

Step 1: Formulate Hypotheses:

- Null Hypothesis $H_0 : p = 0.20$ (20% of passengers travel in first class).
- Alternative Hypothesis $H_1 : p \neq 0.20$ (Two-tailed test).

Step 2: Sample Proportion & Standard Error:

- Sample size $n = 400$
- Successes $x = 70 \implies \hat{p} = \frac{70}{400} = 0.175$
- Hypothesized proportion $p_0 = 0.20 \implies q_0 = 1 - 0.20 = 0.80$

$$\sigma_p = \sqrt{\frac{p_0 \cdot q_0}{n}} = \sqrt{\frac{0.20 \times 0.80}{400}} = \sqrt{\frac{0.16}{400}} = \sqrt{0.0004} = 0.02$$

Step 3: Compute Z_{cal} :

$$Z_{\text{cal}} = \frac{\hat{p} - p_0}{\sigma_p} = \frac{0.175 - 0.20}{0.02} = \frac{-0.025}{0.02} = -1.25$$

Step 4: Decision:

At $\alpha = 0.05$ (two-tailed), critical $Z = \pm 1.96$.

$$|Z_{\text{cal}}| = 1.25 < 1.96 \implies \text{Fail to Reject } H_0$$

Conclusion: We cannot reject the null hypothesis. The observed difference (17.5% vs 20%) is attributable to chance sampling variation at the 5% level.

SOLVED EXAM PROBLEM 3.3: SMALL SAMPLE STUDENT'S TTT-TEST (MAKEUP APR 2025, Q6B, 10 MARKS)

Problem Statement: The average adult male height in a certain country is 170 cm. We suspect that the men in a certain city in that country might have a different average height due to environmental factors. We pick a random sample of size 9 from the adult males in the city and obtain the following values for their heights (in cm):

176.2, 157.9, 160.1, 180.9, 165.1, 167.2, 162.9, 155.7, 166.2.

Test whether average male height can be 170 cm. Consider significance level to be 5%.

Step 1: State Hypotheses:

- $H_0 : \mu = 170$ cm (Average male height in the city is 170 cm).
- $H_1 : \mu \neq 170$ cm (Two-tailed test).

Step 2: Compute Sample Mean ($\bar{X}$) and Sample Standard Deviation (s):

$$\sum X = 176.2 + 157.9 + 160.1 + 180.9 + 165.1 + 167.2 + 162.9 + 155.7 + 166.2$$

$$\bar{X} = \frac{1492.2}{9} = 165.80 \text{ cm}$$

Deviations ($X_i - \bar{X}$) and squared deviations ($(X_i - \bar{X})^2$):

- $176.2 - 165.8 = +10.4 \implies (10.4)^2 = 108.16$
- $157.9 - 165.8 = -7.9 \implies (-7.9)^2 = 62.41$
- $160.1 - 165.8 = -5.7 \implies (-5.7)^2 = 32.49$
- $180.9 - 165.8 = +15.1 \implies (15.1)^2 = 228.01$
- $165.1 - 165.8 = -0.7 \implies (-0.7)^2 = 0.49$
- $167.2 - 165.8 = +1.4 \implies (1.4)^2 = 1.96$
- $162.9 - 165.8 = -2.9 \implies (-2.9)^2 = 8.41$
- $155.7 - 165.8 = -10.1 \implies (-10.1)^2 = 102.01$
- $166.2 - 165.8 = +0.4 \implies (0.4)^2 = 0.16$

$$\sum (X_i - \bar{X})^2 = 108.16 + 62.41 + 32.49 + 228.01 + 0.49 + 1.96 + 8.41 + 102.01 + 0.16$$

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}} = \sqrt{\frac{544.10}{8}} = \sqrt{68.0125} \approx 8.2470 \text{ cm}$$

Step 3: Calculate Student's t Statistic:

$$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} = \frac{165.80 - 170}{8.2470 / \sqrt{9}}$$

SOLVED PROBLEM 3.4: CHI-SQUARE (χ^2) TEST OF INDEPENDENCE ON 2×2 TABLE

Problem Statement: A survey evaluated 100 software repositories to test whether defect rates depend on development methodology (Agile vs. Waterfall).

The observed counts are:

Methodology	High Defect (> 10)	Low Defect (≤ 10)	Total
Agile	15	35	50
Waterfall	30	20	50
Total	45	55	100

Test at the 5% level ($\alpha = 0.05$) whether defect rate is independent of development methodology.

Step 1: Compute Expected Frequencies ($E = \frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}}$):

- $E_{11} = \frac{50 \times 45}{100} = 22.5$
- $E_{12} = \frac{50 \times 55}{100} = 27.5$
- $E_{21} = \frac{50 \times 45}{100} = 22.5$
- $E_{22} = \frac{50 \times 55}{100} = 27.5$

Step 2: Calculate χ^2_{cal} :

$$\chi^2 = \frac{(15 - 22.5)^2}{22.5} + \frac{(35 - 27.5)^2}{27.5} + \frac{(30 - 22.5)^2}{22.5} + \frac{(20 - 27.5)^2}{27.5}$$

$$\chi^2 = \frac{(-7.5)^2}{22.5} + \frac{(7.5)^2}{27.5} + \frac{(7.5)^2}{22.5} + \frac{(-7.5)^2}{27.5} = \frac{56.25}{22.5} + \frac{56.25}{27.5} + \frac{56.25}{22.5} + \frac{56.25}{27.5}$$

$$\chi^2 = 2.5000 + 2.0455 + 2.5000 + 2.0455 = \mathbf{9.0910}$$

Step 3: Degrees of Freedom & Critical Value:

$$df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$$

Critical $\chi^2_{0.05,1} = \mathbf{3.841}$.

$$\chi^2_{\text{cal}} = \mathbf{9.0910} > \mathbf{3.841} \implies \text{Reject } H_0$$

Conclusion: Reject H_0 . There is a statistically significant association between development methodology and defect rates ($p < 0.01$).

INTERACTIVE STATISTICAL STUDIO

Hypothesis Testing & Parametric Test SimulatorZ-Test &
FPCStudent's t-
Test

Test hypotheses for population proportions or finite population means with automatic Finite Population Correction (FPC):

$$Z_{\text{cal}} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}, \quad Z_{\text{FPC}} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}}$$

Exam Presets: **Passenger First-Class (SEE 2025: p0=0.20, n=400, x=70)**

Zoo Field Trip (Makeup 2025: p0=0.85, n=50, x=39)

Labor Turnover FPC (SEE 2025: N=20, n=5, μ=320)

Hypothesized Proportion

Sample Size (n):

Observed Successes (x):

(p₀):

400

70

0.20

Significance Level (α):

α = 0.05 (Z_{crit} = ±: ▼)

CALCULATED Z-STATISTIC

-1.2500p_{hat} = 0.1750, SE = 0.0200

STATISTICAL DECISION

FAIL TO REJECT H₀ ✓

| -1.2500 | < 1.96 (Consistent with null hypothesis)

10. Quick Revision Sheet

Test / Formula	When to Use	Test Statistic Formula	Critical Cut-off ($\alpha = 0.05$, two-tailed)
Z-Test (Mean)	$n \geq 30$ or known σ . (Apply FPC if $n/N > 0.05$)	$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}} \cdot \text{FPC}}$	± 1.96
Z-Test (Proportion)	Categorical success rate ($n \geq 30$)	$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$	± 1.96
Student's t-Test	Small sample $n < 30$, unknown σ , normal population	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$	Read from t -table at $df = n - 1$
Chi-Square (χ^2)	Test of independence on contingency tables ($r \times c$)	$\chi^2 = \sum \frac{(O-E)^2}{E}, \quad df = (r-1)(c-1)$	3.841 for $df = 1$
ANOVA (F-Test)	Testing equality of $k \geq 3$ population means	$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}}$	Read from F -table at $(k-1, N-k)$

Authentic Exam Archive & Model Solutions

■ ARCHIVE

The following authentic questions have been transcribed directly from official department examination papers in notes/rmipr/practice/:

Ramaiah Autonomous Examination · CIE-2 (Dec 22, 2025) Q1.a [8 Marks]

"Compare and contrast the observation method and interview method of primary data collection."

Official Model Solution:

Comparison Dimension	Observation Method	Interview Method
Mode of Data Capture	Direct monitoring and recording of behavior without asking respondents.	Oral verbal interaction (face-to-face or telephone) with direct questioning.
Subjectivity & Bias	Eliminates respondent self-report bias; observer bias is controlled via structured observation protocols.	Susceptible to social desirability bias from respondents and interviewer tone/probing bias.
Depth of Internal Insight	Limited strictly to observable external actions; cannot probe underlying thoughts, past history, or private motives.	Deep psychological probing into past experiences, motivations, emotional nuances, and attitudes.
Respondent Cooperation	Does not require active respondent willingness; ideal for automated sensors, animals, or unconscious patients.	Requires full active cooperation, availability, and articulacy of the interviewee.
Cost & Time	Expensive per unit of data captured due to prolonged idle waiting for spontaneous events.	Higher throughput per appointment; scheduling overhead and geographic travel costs can be substantial.

Ramaiah Autonomous Examination · CIE-2 (Dec 22, 2025) Q3.b [7 Marks]

"Explain the steps involved in sample design."

Official Model Solution: Developing an authentic sample design follows six structured steps (C.R. Kothari Ch. 4):

1. **Type of Universe (Target Population):** Define whether the universe of study is *finite* (e.g. 5,000 enrolled students) or *infinite* (e.g. continuous network packet streams).
2. **Sampling Unit:** Decide the basic operational unit to be sampled before selecting items (geographical units like states/districts, institutional units like schools, or individual persons).
3. **Source List (Sampling Frame):** Secure the authoritative register, roster, or directory containing every sampling unit from which the sample is drawn (e.g. university registrar enrollment database).
4. **Sample Size (n):** Calculate the mathematically optimal number of items ensuring adequate precision, confidence level, and acceptable margin of error within budget constraints.
5. **Budgetary Constraints:** Balance statistical variance reduction against real-world limitations of cost, compute time, and field investigator manpower.
6. **Sampling Procedure:** Select the exact probability sampling method (Simple Random, Stratified, Systematic, Cluster, or Multi-stage) that minimizes sampling error.

Ramaiah Autonomous Examination · SEE (Feb/Mar 2025) Q5.b [10 Marks]

"Suppose we are interested in a population of 20 industrial units of the same size, all of which are experiencing excessive labour turnover problems. The past records show that the mean annual turnover is 320 employees, with a standard deviation of 75 employees. A sample of 5 of these units is taken at random which gives a mean turnover of 300 employees. Is the sample mean consistent with the population mean? Test at 5% level."

Official Model Solution:

See **Solved Exam Problem 3.1** above for the full step-by-step derivation:

- $N = 20, n = 5 \implies \frac{n}{N} = 0.25 > 0.05 \implies \text{FPC factor required!}$
- $\text{FPC} = \sqrt{\frac{20-5}{20-1}} = \sqrt{\frac{15}{19}} \approx 0.8885$
- $\sigma_{\bar{X}} = \left(\frac{75}{\sqrt{5}}\right) \times 0.8885 = 29.8020$
- $Z_{\text{cal}} = \frac{300-320}{29.8020} = -0.6711$
- Critical value for two-tailed test at $\alpha = 0.05$ is $Z_{\text{crit}} = \pm 1.96$.
- Since $|-0.6711| < 1.96$, **Fail to reject H_0** . The sample turnover of 300 is consistent with the population mean of 320.

Ramaiah Autonomous Examination · SEE (Feb/Mar 2025) Q6.b [10 Marks]

"The null hypothesis is that 20 per cent of passengers go in first class, but management recognizes the possibility that this percentage could be more or less. A random sample of 400 passengers includes 70 holding first class tickets. Can the null hypothesis be rejected at the 5% level of significance?"

Official Model Solution:

- $H_0 : p = 0.20, \quad H_1 : p \neq 0.20$
- $n = 400, x = 70 \implies \hat{p} = \frac{70}{400} = 0.175$
- $\sigma_p = \sqrt{\frac{0.20 \times 0.80}{400}} = \sqrt{0.0004} = 0.02$
- $Z_{\text{cal}} = \frac{0.175-0.20}{0.02} = -1.25$
- $|Z_{\text{cal}}| = 1.25 < 1.96 \implies \text{Fail to reject } H_0.$

Practice Problems with Hidden Step-by-Step Solutions

PRACTICE 1 · ZOO TRIP PROPORTION TEST

A teacher believes 85% of students want a zoo trip ($p_0 = 0.85$). A sample of 50 students yields 39 wanting to go. Test at $\alpha = 0.05$ whether the proportion differs from 85%.

Practice Problem:

Show Step-by-Step Solution

Ans

$$p_0 = 0.85, \quad q_0 = 0.15, \quad n = 50, \quad x = 39 \implies \hat{p} = 0.78$$

$$\sigma_p = \sqrt{\frac{0.85 \times 0.15}{50}} = \sqrt{0.00255} \approx 0.0505$$

$$Z_{\text{cal}} = \frac{0.78 - 0.85}{0.0505} = -1.3862$$

Critical value at $\alpha = 0.05$ is ± 1.96 . Since $|-1.3862| < 1.96$, **Fail to reject** H_0 . (Verified in audit/verify/rmipr/verify_rmipr_u3.py).

PRACTICE 2 · STUDENT'S t -TEST ON HEIGHTS

A sample of 9 adult males yields heights: 176.2, 157.9, 160.1, 180.9, 165.1, 167.2, 162.9, 155.7, 166.2. Test at $\alpha = 0.05$ whether the population mean is 170 cm.

 Practice Problem:

Show Step-by-Step Solution

Ans

Sample mean $\bar{X} = 165.80$ cm, sample standard deviation $s = 8.2470$ cm.

$$SE = \frac{s}{\sqrt{n}} = \frac{8.2470}{3} = 2.7490$$

$$t_{\text{cal}} = \frac{165.80 - 170.0}{2.7490} = -1.5278$$

At $df = 8$ and $\alpha = 0.05$, critical $t_{0.025,8} = 2.306$. Since $|-1.5278| < 2.306$, **Fail to reject** H_0 . Average male height can be 170 cm. (Verified in audit/verify/rmipr/verify_rmipr_u3.py).

PRACTICE 3 · CHI-SQUARE TEST OF INDEPENDENCE

In a 2×2 contingency table with observed counts $O = [15, 35, 30, 20]$ and equal marginal totals (50, 50, 45, 55), compute the calculated χ^2 value and evaluate at $\alpha = 0.05$.

 Practice Problem:

Show Step-by-Step Solution

Ans

Expected frequencies: $E = [22.5, 27.5, 22.5, 27.5]$.

$$\chi^2 = \frac{(15 - 22.5)^2}{22.5} + \frac{(35 - 27.5)^2}{27.5} + \frac{(30 - 22.5)^2}{22.5} + \frac{(20 - 27.5)^2}{27.5} = 2.5$$

At $df = 1$, critical $\chi_{0.05,1}^2 = 3.841$. Since $9.091 > 3.841$, **Reject** H_0 . (Verified in audit/verify/rmipr/verify_rmipr_u3.py).

PRACTICE 4 · STEVENS' 4 MEASUREMENT SCALES

Classify each measurement scale: (a) CPU Temperature in Kelvin, (b) Server rack IDs, (c) Student letter grades (A, B, C, D).

 Practice Problem:

Show Step-by-Step Solution

Ans

- (a) **Temperature in Kelvin: Ratio Scale** (Possesses an absolute true zero point where thermal molecular motion ceases).
- (b) **Server rack IDs: Nominal Scale** (Qualitative identifiers used strictly for labeling; arithmetic operations have no physical meaning).
- (c) **Letter grades: Ordinal Scale** (Rank ordered, but intervals between grades are not strictly equidistant).

PRACTICE 5 · TYPE I VS TYPE II ERROR TRADEOFF

Explain what happens to the probability of committing a Type II Error (β) when an investigator reduces the significance level (α) from 0.05 to 0.01.

 Practice Problem:

Show Step-by-Step Solution

Ans

For a fixed sample size n , α and β are inversely related. Reducing α from 0.05 to 0.01 makes the critical boundary more stringent, decreasing the probability of a false alarm (Type I error). However, this expands the non-rejection region, **increasing the probability of committing a Type II Error (β)** and lowering statistical test power ($1 - \beta$).

PRACTICE 6 · ANOVA F-RATIO INTERPRETATION

In an ANOVA test with $k = 4$ groups and total $N = 40$ units, what are the numerator and denominator degrees of freedom for the F -statistic?

 Practice Problem:

Show Step-by-Step Solution

Ans

- Numerator $df_{\text{Between}} = k - 1 = 4 - 1 = 3$
- Denominator $df_{\text{Within}} = N - k = 40 - 4 = 36$
- Total $df = N - 1 = 39$. Sum check: $3 + 36 = 39$.

12. Prescribed References

OFFICIAL COURSE TEXTBOOKS

1. **C.R. Kothari and Gourav Garg**, *Research Methodology: Methods and Techniques*, 4th Edition, New Age International Publishers. (Chapters 5, 6, 8 & 9).
2. **David V. Thiel**, *Research Methods for Engineers*, Cambridge University Press, ISBN 978-1-107-03488-4. (Chapters 6 & 7).
3. **Dipankar Deb, Rajeeb Dey, Valentina E. Balas**, *Engineering Research Methodology*, Springer Intelligent Systems Reference Library, ISBN 978-981-13-2946-3.

AI-assisted study notes written from the official syllabus. Verify against the prescribed textbook and your faculty's notes before the exam.